Abhishek Ningappa Walikar

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Profile

M.Sc. Computer Science student at Universität Paderborn with professional experience in Python, machine learning, data workflows, and software development. Hands-on experience includes LLM applications, RAG, LangGraph-based agentic workflows, explainable AI, model evaluation, and reproducible ML experimentation. Experienced in developing AI prototypes, quantitatively evaluating alternative approaches, and strengthening solutions through automated testing, Docker, and CI/CD. Interested in applied Data & AI, AI Engineering, and translating technical AI capabilities into practical business value.

Education

04/2025 – Present **M.Sc. in Computer Science, Universität Paderborn**
Paderborn, Germany | Expected 03/2028 | Current Grade: 2.6

- Relevant coursework: Machine Learning for Face Biometrics, Explainable Artificial Intelligence, Data-Driven Engineering (1.3), Data-Driven Innovation (2.0).

01/2021 – 08/2022 **Master of Computer Applications (MCA), Karnatak University**
India

08/2017 – 10/2020 **B.Sc. Computer Science, JSS SMI UG & PG Studies**
India

Professional Experience

04/2023 – 12/2024 **Software Engineer – Python, Data & ML**
CABCS Group India Technologies Pvt. Ltd., Bangalore, India

- Developed and maintained Python-based software and data-processing workflows for extraction, cleansing, validation, and preparation across multiple datasets, supporting repeatable analytics and ML workloads.
- Fine-tuned and evaluated a Hugging Face Transformer with PyTorch for automated text classification, achieving **96%+ classification accuracy** on a self-collected and curated dataset.
- Built reusable ML experimentation pipelines using scikit-learn and TensorFlow across different datasets and model configurations, supporting structured validation, benchmarking, and reproducible analysis.

08/2022 – 02/2023 **Full-Stack Developer Intern**
KodNest India Pvt. Ltd., Bangalore, India

- Developed 10+ Python application features and integrated/tested 6+ REST APIs, gaining practical experience with backend logic, databases, application testing, and reliable data flow.

Key Projects

01/2025 – 04/2026 **LLM-Based System Context Analysis Assistant**
Applied LLM & Software Engineering Project | Universität Paderborn
GitHub: github.com/abhiwalikar1999/LLM-Based-System-Context-Analysis-Assistant

- Developed a Python LLM system with structured outputs, Pydantic validation, custom embedding-based RAG, LangGraph self-validation, and three interactive analysis views.
- Built a quantitative evaluation harness achieving **0.88 Entity F1** and **0.82 Relationship F1**; agentic validation improved Entity F1 from **0.879 to 0.924**. Added **65 pytest tests**, Docker, GitHub Actions CI/CD, and live deployment.

04/2026 – 10/2026 **Explainable Knowledge-Graph Classification**
Machine Learning & Explainable AI Project | Universität Paderborn
GitHub: github.com/abhiwalikar1999/explainable-kg-classification

- Developed a leakage-safe AI classification pipeline, identifying and removing **79 residual label-leakage edges** and **414 of 479 identity-memorising features**; added automated leakage tests and nested cross-validation.
- Engineered approximately **1,150 human-readable graph features** and compared models across approximately **8,800 nested-CV fits**, reaching **89.8% CV accuracy** with LightGBM and analysing results with SHAP and model-ablation methods.

10/2025 – 08/2026 **Machine Unlearning for Face Recognition – Model Evaluation & Reliability**
Research Project | Deep Learning & Computer Vision | Universität Paderborn

- Co-developed an adversarial-probing evaluation method and evaluated it on a **5.82M-image dataset** across 3 machine-unlearning techniques and 7 model configurations, producing stronger separation in **40 of 41** completed single-identity configurations.
- Optimized adversarial-search settings, reducing full-run runtime from **1,215 s to 690 s** (approximately **43%**) while changing the evaluation score by less than **0.001**.

Technical Skills

Programming	Python, SQL
AI / LLM	LLM API integration, RAG, LangGraph, prompt engineering, Pydantic
ML / Data	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LightGBM, pandas, NumPy, SHAP
Engineering	Git/GitHub, pytest, Docker, GitHub Actions CI/CD, REST APIs, Linux, Jupyter/Colab

Languages

English	Fluent (C1)
German	A2 completed, learning B1